INCH - POUND

MIL-DTL-39024D
12 December 2019
SUPERSEDING
MIL-DTL-39024C
15 November 2014

DETAIL SPECIFICATION

# JACK, TIP (TEST POINT, PANEL OR PRINTED WIRING TYPE), GENERAL SPECIFICATION FOR

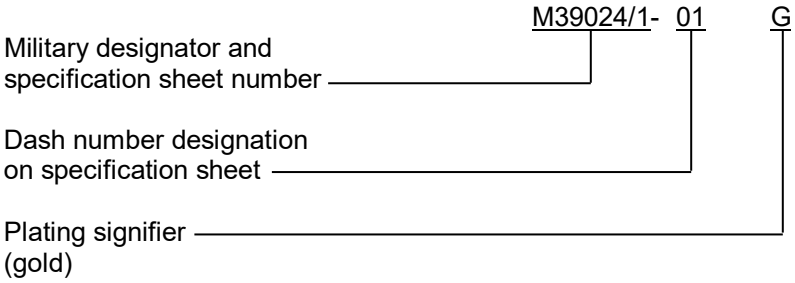
This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the general requirements for panel or printed wiring type test point connectors for use in electrical and electronic equipment (see 6.1). As used in this specification, the term “connectors” includes tip jacks.

1.2 Classification. Connectors covered by this specification are single-contact or multiple-contact connectors, as specified (see 3.1).

1.2.1 Part or Identifying Number (PIN). The military PIN consists of the letter “M” followed by the basic number of specification sheet, and an assigned dash number (see 3.1), as shown in the following:




## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, Attn: VAI, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to RFConnector@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.



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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-DTL-55330 - Connectors, Electrical and Fiber Optic, Packaging of.

(See supplement 1 for associated specification sheets.)

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

DEPARTMENT OF DEFENSE

MIL-STD-202 - Electronic and Electrical Component Parts.
MIL-STD-202-107 - Thermal shock .
MIL-STD-202-208 - Solderability.
MIL-STD-202-302 - Insulation resistance.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Copies of this document are available online at https://assist.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) INTERNATIONAL

ASTM B16/B16M - Rod, Brass, Free-cutting, Bar and Shapes for Use in Screw Machines.
ASTM B139/B139M - Rod, Phosphor Bronze, Bar, and Shapes.
ASTM B194 - Copper Beryllium Alloy Plate, Sheet, Strip and Rolled Bar.
ASTM B196/B196M - Rod and Bar, Copper-Beryllium Alloy.
ASTM B197/B197M - Wire, Alloy Copper-Beryllium.
ASTM A342/A342M - Materials, Feebly Magnetic, Permeability Of.
ASTM B488 - Electrodeposited Coatings of Gold for Engineering Uses.
ASTM D635 - Plastics in a Horizontal Position, Rate of Burning and/or Extent and Time of Burning of.
ASTM B740 - Copper-Nickel-Tin Spinodal Alloy Strip.
ASTM A967 - Chemical Passivation Treatments for Stainless Steel Parts.
ASTM D2863 - Plastics, Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of (Oxygen Index).
ASTM D4894 - Polytetraflouroethylene (PTFE) Granular Molding and Ram Extrusion Materials.
ASTM D5948 - Compounds, Molding, Thermosetting.

(Copies of these documents are available at http://www.astm.org.)



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ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

EIA/ECA-364-06 - Contact Resistance Test Procedure for Electrical Connectors.
EIA/ECA-364-20 - Withstanding Voltage Test Procedure for Electrical Connectors, Sockets and Coaxial Contacts.
EIA/ECA-364-21 - Insulation Resistance Test Procedure for Electrical Connectors, Sockets, and Coaxial Contacts.
EIA/ECA-364-26 - Salt Spray Test Procedure for Electrical Connectors, Contacts and Sockets.
EIA/ECA-364-27 - Mechanical Shock (Specified Pulse) Test Procedure for Electrical Connectors and Sockets.
EIA/ECA-364-31 - Humidity Test Procedure for Electrical Connectors and Sockets.
EIA/ECA-364-32 - Thermal Shock (Temperature Cycling) Test Procedure for Electrical Connectors and Sockets.
EIA/ECA-364-38 - Cable Pull-Out Test Procedure for Electrical Connectors.

(Copies of these documents are available online at http://eciaonline.org.)

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELEC CIRCUITS (IPC)

IPC 2221 - Printed Boards Design, Generic Standard for.
J-STD-004 - Soldering Fluxes, Requirements for.
J-STD-005 - Soldering Pastes, Requirements for.
J-STD-006 - Electronic Grade Solder Alloys and Fluxed and Non-fluxed Solid Solders for Electronic Soldering, Applications Requirements for.

(Application for copies should be addressed to the Institute for Interconnecting and Packaging Elec Circuits, 2215 Sanders Road, Northbrook, Il 60062.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC17025 - General requirements for the competence of testing and calibration laboratories.

(Copies of these documents are available online at http://www.iso.ch.)

AMERICAN SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS 2418 - Copper Plating.
SAE AMS-P-81728 - Plating, Tin-Lead (Electrodeposited).
SAE AMS QQ-P-35 - Passivation Treatment for Corrosion-Resistant Steel.
SAE AMS-QQ-N-290 - Nickel Plating (Electrodeposited).
SAE AMS -STD-595 - Colors used in Government Procurement.

(Copies of these documents are available at http://www.sae.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

HP-3 - Electrical and Electronic PTFE (Polytetraflouroethylene) Insulated High Temperature Hook-up Wire; Types ET (250 Volts), E (600 Volt), and EE (1000 Volts).

(Copies of these documents are available online at http://www.nema.org.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or



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other informational services.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

3.3 Materials. Materials shall be as specified herein; however, when a definite material is not specified, a material shall be used which will enable the connectors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Metal parts.

3.3.1.1 Nonmagnetic materials. All parts shall be made from materials which are classified as nonmagnetic.

3.3.1.2 Dissimilar metals. Where dissimilar materials are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. Dissimilar metals shall be as defined in MIL-STD-889. Dissimilar metals such as brass, copper, or steel (except corrosion-resisting steel, passivated in accordance with SAE AMS QQ-P-35 or ASTM A967) shall not be used in intimate contact with aluminum or aluminum alloy.

3.3.1.3 Metals and finishes. All exposed metals parts, except electric contacts, terminals, and corrosion-resisting steel parts shall be nickel-plated in accordance with class 1, type 2, of SAE AMS-QQ-N-290. Steel parts shall be passivated in accordance with SAE AMS QQ-P-35 or ASTM A967.

3.3.1.4 Contacts. Contact springs and tabs shall be made of beryllium copper in accordance with ASTM B194, ASTM B196/B196M, or ASTM B197/B197M; or when specified (see 3.1) phosphor bronze in accordance with ASTM B139/B139M or copper nickel alloy C72900 in accordance with ASTM B740. The contact springs and tabs, and that portion of the body that provides a normal force to the test probe shall be gold plated in accordance with ASTM B488, type 3, code C, class 1.27, over an underplate either of copper, in accordance with SAE AMS 2418; 0.0002 inch minimum thickness, or nickel in accordance with class 2 of SAE AMS-QQ-N-290, 0.00003 to 0.00015 inch thickness.

3.3.1.5 Terminals. Soldering terminals shall be made of a copper alloy material and shall be plated in one of the following manners:

a. Gold plated in accordance with ASTM B488, type 3, code B or C, class 0.50 (0.00002 in) over a copper underplate in accordance with SAE AMS 2418, 0.0002 inch minimum.
b. Gold plated in accordance with ASTM B488, type 3, code B or C, class 0.50 (0.00002 in) over a nickel underplate in accordance with SAE AMS-QQ-N-290, class 2, 0.00003 to 0.00015 inch thickness.

c. Tin lead plated in accordance with SAE AMS-P-81728, 50-70 percent tin 0.0001-inch thick minimum over nickel in accordance with SAE AMS-QQ-N-290, class 2, 0.00003 to 0.00015 inch thick. For devices used in printed wiring applications.



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d. Tin lead plated in accordance with SAE AMS-P-81728, 50 to 95 percent tin 0.0001-inch thick minimum over nickel in accordance with SAE AMS-QQ-N-290, class 2, 0.00003 to 0.00015 inch thick. For devices used in non-printed wiring applications.

e. Solder coated in accordance with J-STD-004, J-STD-005, and J-STD-006, SN60, 0.0003 inch thick over copper accordance with SAE AMS 2418, 0.0002 inch minimum thickness.

f. Solder coated in accordance with J-STD-004, J-STD-005, and J-STD-006, SN60, 0.0003 inch thick over nickel accordance with SAE AMS-QQ-N-290, class 2, 0.00003 to 0.00015 inch thick.

NOTE: Those parts with gold plated solder terminals shall be identified by a suffix G on the PIN, for example, M39024/18-07G. For replacement purposes, the Government will stock and issue only those parts having gold plated solder terminals.

3.3.2 Plastic parts. Plastic parts shall be made of type GDI-30F or type SDG-F of ASTM D5948, or, when specified (see 3.1), polytetrafluoroethane of ASTM D4894. When applicable, insulation colors shall conform to SAE AMS -STD-595, as specified (see 3.1).

3.3.2.1 Flammability. Plastic materials shall be limited to those certified by their manufacturers as self-extinguishing in accordance with ASTM D635 and ASTM D2863.

3.3.3 Threaded parts. Screw threads for threaded parts shall conform to FED-STD-H28, and shall be as specified (see 3.1).

3.4 Design, construction, and physical dimensions. Connectors shall be of the design, construction, and physical dimensions specified (see 3.1). Unless otherwise specified (see 3.1), connectors shall have an operating temperature range of -65°C to +125°C.

3.4.1 Contacts. Contacts shall be closed-entry design. The closed-entry feature shall be an integral part of the socket contact. The socket contact shall provide the spring action for maintaining the contact pressure between the test probe and socket. Contacts shall accept test probes 0.040 ± 0.001 or 0.080 ± 0.001 inch in diameter, as specified (see 3.1).

3.4.1.1 Contact identification. Contact positions on multiple-contact connectors shall be permanently identified by legible letters or numerals, as specified (see 3.1), molded or stamped on the front and rear face of the connector body. Marking shall be arranged to avoid confusion between contacts.

3.4.1.2 Contact arrangement. The center-to-center distance between contacts shall be as specified (see 3.1).

3.4.1.3 Contact finish. Contact finish shall be smooth, free of shear lines, tear out or scratches, and shall show no signs of porosity or surface cracks.

3.4.1.4 Contact current rating. The current rating of contacts shall be as shown in table I, as specified (see 3.1).

TABLE I. Contact current rating.

| Contact diameter (inches) | Maximum current rating (amperes) |
|---|---|
| 0.040 | 3 |
| 0.080 | 5 |

3.4.2 Mounting hardware. Screws, clamps, brackets, or similar means for mounting the connectors



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shall be furnished, when specified (see 3.1).

3.5 Performance.

3.5.1 Permeability. When connectors are tested as specified in 4.6.3, the permeability shall be less than 2.0 Mu.

3.5.2 Contact resistance. When connectors are tested as specified in 4.6.4, the voltage drop shall be not greater than 20 millivolts, unless otherwise specified (see 3.1).

3.5.3 Insulation resistance. When measured as specified in 4.6.5, the insulation resistance shall be 5,000 megohms or greater.

3.5.4 Dielectric withstanding voltage. When tested as specified in 4.6.6, connectors shall withstand the test voltage specified (see 3.1) with no evidence of flashover.

3.5.5 Insertion and withdrawal forces. When connectors are tested as specified in 4.6.7, the forces required to insert and withdraw the test probes (see 4.6.1.1) shall be as specified (see 3.1).

3.5.6 Resistance to test-probe damage. When tested as specified in 4.6.8, socket contacts and connectors shall be capable of withstanding the bending moment and depth of test probe insertion with no visual evidence of physical damage.

3.5.7 Thermal shock. When tested as specified in 4.6.9, connectors shall show no visual evidence of physical damage.

3.5.8 Shock (specified pulse). When tested as specified in 4.6.10, connectors shall show no visual evidence of mechanical damage, loosening of parts, rupturing of dielectric materials, or loss of electrical continuity in excess of 10 microseconds.

3.5.9 Vibration. When connectors are tested as specified in 4.6.11, there shall be no loss of electrical continuity in excess of 10 microseconds during the test. After the test, connectors shall meet the following requirements:

| | |
|---|---|
| Visual examination: | There shall be no cracking, breaking, or loosening of parts or other evidence of physical damage. |
| Insertion and withdrawal forces: | Shall be as specified in 3.5.5. |
| Contact resistance: | Shall be as specified in 3.5.2, unless otherwise specified (see 3.1). |

3.5.10 Humidity (steady state). When tested as specified in 4.6.12, connectors shall show no visual evidence of physical damage and the insulation resistance shall exceed 100 megohms after the test and shall exceed 1,000 megohms after a 24-hour drying period.

3.5.11 Salt spray (corrosion). When connectors are tested specified in 4.6.13, there shall be no exposure of the basis metal. After the test, the contact resistance shall be as specified in 3.5.2, unless otherwise specified (see 3.1).

3.5.12 Durability. When tested as specified in 4.6.14, connectors shall show no visual evidence of electrical or mechanical damage and shall meet the following requirements:

| | |
|---|---|
| Insertion and withdrawal forces: | Shall be as specified in 3.5.5. |
| Contact resistance: | Shall be as specified in 3.5.2, unless otherwise specified (see 3.1). |

3.5.13 Solderability. Solderable surfaces shall meet the requirements of MIL-STD-202-208 for



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solderability after 8 hours steam aging.

3.6 Identification marking. Connectors shall be marked in accordance with method I of MIL-STD-1285, with the complete PIN, manufacturer's name or trademark, and when specified (see 3.1), the date and source code. Marking shall be located on the connector so that it will be visible after installation, unless this is impossible because of space limitations. In such cases, required marking shall be applied to an envelope, bag, box or other intermediate container suitable for shelf storage.

3.7 Workmanship. Connectors shall be processed in such a manner as to be uniform in quality and shall be free from pits, cracks, rough edges, and other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with NCSL Z540.3.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4 First article inspection. First article inspection shall be performed on connectors of each type, size, and terminal style to be delivered (if applicable) when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4.1, 4.4.2, 4.4.3 and the tests of 4.5.1 through 4.5.6.

4.4.1 Sample size. Twenty-four connectors of each type, size, and terminal style (if applicable) shall be subjected to qualification inspection.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table II, in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall then be divided equally into four groups of six units each, and subjected to the inspections for their particular group.

4.4.3 Failures. One or more failures shall be cause for refusal of the lot.

4.5 Conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.5.2 Inspection lot. An inspection lot, as far as practicable, shall consist of all connectors of the same basic type, size, and terminal style (if applicable), produced under essentially the same conditions, and offered for inspection at one time. When a connector has different colors of insulation, an inspection lot may include any quantity and variety of colors provided that the units are otherwise mechanically and dimensionally identical.

4.5.2.1 Group A inspection. Group A inspection shall consist of the examination and test specified in table III, and shall be made on the same set of sample units, in the order shown.



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4.5.2.1.1 Sampling plan (group A). Table III tests shall be performed on a production lot basis. Samples shall be selected in accordance with table IV. If one or more defects are found, the lot shall be screened for those particular defects and the defects shall be removed. A new sample of parts shall be selected in accordance with table IV and all group A tests again performed. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE II. First article inspection.

| Inspection | Requirement paragraph | Test paragraph |
|---|---|---|
| Group I (all specimens) 1/ | | |
| Visual and mechanical examination | 3.1, 3.3, 3.4, 3.6 and 3.7 | 4.6.2 |
| Permeability | 3.5.1 | 4.6.3 |
| Group II (6 specimens) | | |
| Contact resistance | 3.5.2 | 4.6.4 |
| Insulation resistance | 3.5.3 | 4.6.5 |
| Dielectric withstanding voltage | 3.5.4 | 4.6.6 |
| Insertion and withdrawal forces | 3.5.5 | 4.6.7 |
| Resistance to test-probe damage | 3.5.6 | 4.6.8 |
| Group III (6 specimens) | | |
| Thermal shock | 3.5.7 | 4.6.9 |
| Shock (specified pulse) | 3.5.8 | 4.6.10 |
| Vibration | 3.5.9 | 4.6.11 |
| Humidity (steady state) | 3.5.10 | 4.6.12 |
| Group IV (6 specimens) | | |
| Salt spray (corrosion) | 3.5.11 | 4.6.13 |
| Durability | 3.5.12 | 4.6.14 |
| Group V (6 specimens) | | |
| Solderability | 3.5.13 | - - - |

1/ Only 2 samples each for physical dimensions.



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TABLE III. Group A inspection.

| Examination or test | Requirement paragraph | Test paragraph | Sampling procedure |
|---|---|---|---|
| Visual and mechanical examination 1/ | 3.1, 3.3, 3.4, 3.6, and 3.7 | 4.6.2 | See table IV |
| Permeability | 3.5.1 | 4.6.3 | |

1/ Only 2 samples each for physical dimensions.

TABLE IV. Inspection level.

| Lot size | | | Visual and mechanical inspection | |
|---|---|---|---|---|
| | | | Major 1/ | Minor 2/ 3/ |
| 1 | to | 3 | All | all |
| 4 | to | 13 | all | 3 |
| 14 | to | 25 | 13 | 3 |
| 26 | to | 50 | 13 | 5 |
| 51 | to | 90 | 13 | 6 |
| 91 | to | 150 | 13 | 7 |
| 151 | to | 280 | 20 | 10 |
| 281 | to | 500 | 29 | 11 |
| 501 | to | 1,200 | 34 | 15 |
| 1,201 | to | 3,200 | 42 | 18 |
| 3,201 | to | 10,000 | 50 | 22 |
| 10,001 | to | 35,000 | 60 | 29 |
| 35,001 | to | 150,000 | 74 | 29 |
| 150,001 | to | 500,000 | 90 | 29 |
| 500,001 | and | over | 102 | 29 |

1/ Major defect: A major defect is a defect, other than critical, that is likely to results in failure, or to reduce materially the usability of the unit of product for its intended purpose.
2/ Minor defect: A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.
3/ Samples inspected for minor defects may be taken from the same samples inspected for major defects, or may be other samples from the same production lot.



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4.5.2.2 Group B inspection. Group B inspection shall consist of the inspections specified in table V in the order shown and shall be made on sample units which have been subjected to and passed the group A inspection.

4.5.2.2.1 Group B sampling plan. A sample of parts shall be randomly selected in accordance with table VI. If one or more defects are found, the lot shall be screened for that particular defect and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected and subjected to all tests in accordance with table V. If one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE V. Group B inspection.

| Test | Requirement paragraph | Test paragraph |
|---|---|---|
| Contact resistance | 3.5.2 | 4.6.4 |
| Insulation resistance | 3.5.3 | 4.6.5 |
| Dielectric withstanding voltage 1/ | 3.5.4 | 4.6.6 |
| Insertion and withdrawal forces | 3.5.5 | 4.6.7 |
| Resistance to test-probe damage | 3.5.6 | 4.6.8 |

1/ Test at altitude is not required.

TABLE VI. Inspection level.

| Lot size | | | Sample size |
|---|---|---|---|
| 1 | to | 4 | all |
| 5 | to | 50 | 5 |
| 51 | to | 90 | 7 |
| 91 | to | 150 | 11 |
| 151 | to | 280 | 13 |
| 281 | to | 500 | 16 |
| 501 | to | 1,200 | 19 |
| 1,201 | to | 3,200 | 23 |
| 3,201 | to | 10,000 | 29 |
| 10,001 | to | 35,000 | 35 |
| 35,001 | and | over | 40 |

4.5.2.2.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract.



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4.5.3. Group C inspection. Group C inspection shall consist of the tests specified in table VII in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed groups A and B inspections.

4.5.3.1. Sampling plan. Ten sample units of the same basic type shall be selected. No failures will be permitted.

TABLE VII. Group C inspection.

| Test | Requirement paragraph | Test paragraph | Number of sample units to be inspected |
|---|---|---|---|
| Subgroup 1 | | | |
| Dielectric withstanding voltage | 3.5.4 | 4.6.6 | |
| Thermal shock | 3.5.7 | 4.6.9 | 6 |
| Shock (specified pulse) | 3.5.8 | 4.6.10 | |
| Vibration | 3.5.9 | 4.6.11 | |
| Humidity (steady state) | 3.5.10 | 4.6.12 | |
| Subgroup 2 | | | 2 |
| Salt spray (corrosion) | 3.5.11 | 4.6.13 | |
| Durability | 3.5.12 | 4.6.14 | |
| Subgroup 3 | | | 2 |
| Solderability | 3.5.13 | ---------- | |

4.5.3.1.2 Disposition of sample units. Sample units which have been subjected to group C inspection, shall not be delivered on the contract.

4.5.3.1.3 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials and processes, and so forth, and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6 Methods of examination and test.

4.6.1 Preparation for testing. Connectors shall be mounted to appropriate metal panels or printed wiring boards utilizing specified mounting hardware. For electrical tests, terminals shall be wired or soldered with type E wire of NEMA HP-3 (size as specified, see 3.1) to the appropriate panel or to the printed wiring boards conforming to IPC 2221, as appropriate.



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4.6.1.1 Electrical test probes. Probes used for test purposes shall be 0.080 ± 0.001 or 0.040 ± 0.001 inch in diameter after plating, as appropriate, and ½ -inch minimum length, or as specified (see 3.1). Probes shall be brass rod, ½ hard (HO2), Copper alloy UNS No. C36000, in accordance with ASTM B16/B16M, and nickel-plated in accordance with type V (FC), class 2, of SAE AMS-QQ-N-290.

4.6.2 Visual and mechanical examination. Connectors shall be examined to verify that the design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.6, and 3.7).

4.6.3 Permeability (see 3.5.1). The permeability of all metal parts shall be measured with an indicator conforming to ASTM A342/A342M.

4.6.4 Contact resistance (see 3.5.2). Connectors shall be tested in accordance with EIA/ECA-364-6. The following details and exception shall apply.

a. Method of connection - In accordance with figure 1. Measurements shall be made after the temperature of the wire has stabilized.
b. Test current - As specified in table I.
c. Points of measurement - Across each mated test probe and connector between the two points indicated “X”—“X” on figure 1.
d. Number of activations prior to measurement - Not applicable.
e. Number of test activations - Not applicable.
f. Number of measurements per activation - Not applicable.

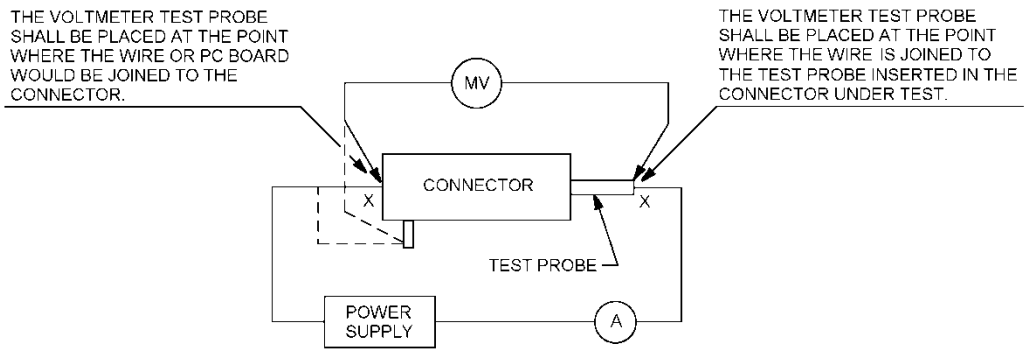


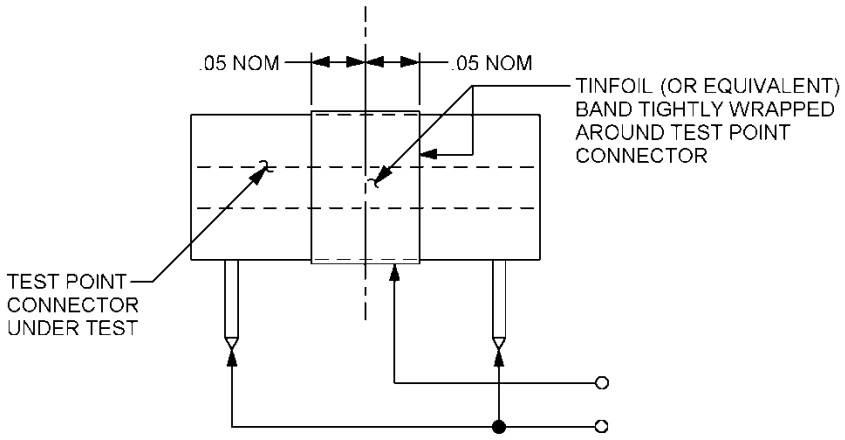

FIGURE 1. Resistance of contacts (potential drop).

4.6.5 Insulation resistance (see 3.5.3). Connectors shall be tested in accordance with method EIA/ECA-364-21 (any number of contacts may be tested simultaneously). The following details and exceptions shall apply:

a. Magnitude of test voltage - 100 volts ± 10 percent. On multiple-contact connectors, the voltage shall be applied between all contacts alternately connected.
b. Special conditions - In accordance with figure 2, when specified (see 3.1).
c. Points of measurement - Between all contacts and the panel or test mount, and between all adjacent contacts.
d. Electrification time - Not to exceed 1 minute.
e. Test condition letter - A of MIL-STD-202-302.



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NOTES:
1. Meters, bridges or other apparatus, when required, shall be incorporated in the circuit.
2. Thickness of the conductive band (e.g., tinfoil) shall be such that it can withstand the applied voltage.

FIGURE 2. Test circuit for dielectric withstanding voltage and insulation resistance tests.

4.6.6 Dielectric withstanding voltage (see 3.5.4). Connectors shall be tested in accordance with EIA/ECA-364-20. The following details shall apply:

a. Special conditions - In accordance with figure 2, when specified (see 3.1).
b. Magnitude of test voltage - As specified (see 3.1).
c. Duration of application of test voltage - For a period of 1 minute, unless otherwise specified (see 3.1).
d. Points of application of test voltage - Between current-carrying parts and ground, unless otherwise specified (see 3.1). On multiple-contact connectors, the voltage shall be supplied between all contacts alternately connected.

4.6.7 Insertion and withdrawal forces (see 3.5.5). A test probe conforming to 4.6.1.1 shall be fully inserted three times into the connector socket contact and withdrawn. The test probe shall again be inserted and insertion and withdrawal forces shall be measured.



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4.6.8 Resistance to test-probe damage (see 3.5.6). A mechanical test probe of hardened steel having a diameter equal to nominal diameter (0.040 ± .001 inch or 0.080 ± .001 inch) mating probe shall be inserted into each socket to 1/8-inch, 3/16-inch, and ¼-inch depths. At each of these depths, measured from the face of the insulator or connector body, a bending moment of 2 inch-pounds ± 10 percent (0.080 test probe) or ½ inch-pound ± 10 percent (0.040 test probe) shall be applied to the probe about the inserted end of the probe and the connector shall be rotated in one direction through 360 degrees, in order that a uniform force is applied to the inside surface of the socket. This test shall be performed with the socket contacts in the insulator or connector body, mounted on a printed wiring board or by panel, as applicable, by the normal means, and the sockets locked and soldered to prevent rotation during the test.
In order to insure uniform test results, the test fixture shown on figure 3 shall be used. Procedures shall be as follows:

a. With the weight in position for the 2 inch-pound or ½ inch-pound moment and size 1 spacer on the pin, insert the pin in a socket while the axis of the socket is in a horizontal position. With the fixture free and unsupported, rotate the socket 360 degrees about a horizontal axis, maintaining the socket in a horizontal position.
b. Repeat with size 2 spacer.
c. Repeat with size 3 spacer.
d. For connectors having a maximum insertion depth of 1/8 inch, only the test using the size 1 spacer(see 4.6.8(a)) is applicable. For connectors having a maximum insertion depth of 3/16 inch, only the tests using sizes 1 and 2 spacers (see 4.6.8 (a) and (b)) are applicable. For connectors having a maximum insertion depth of ¼ inch, tests using all three spacers are required.

After withdrawal of the fixture, the socket shall be visually examined for evidence of physical damage.

4.6.9 Thermal shock (see 3.5.7). Connectors with unwired test probes inserted into contact sockets shall be tested in accordance with EIA/ECA-364-32. The following details shall apply:

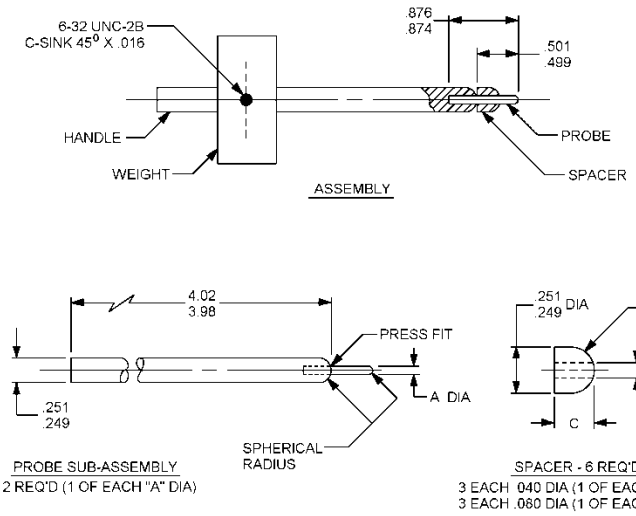
a. Test condition letter – B of MIL-STD-202-107 (or C, when specified (see 3.1).
b. Examination after test – Connectors shall be visually examined for evidence of physical damage.

4.6.10 Shock (specified pulse) (see 3.5.8). Connectors, properly mated, shall be tested in accordance with EIA/ECA-364-27. The following details and exception shall apply:

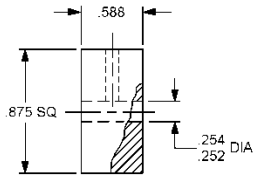
a. Mounting methods - By normal mounting means (see 4.6.1).
b. Test condition letter: - G.
c. Measurements during shock test - Connectors shall be monitored for electrical continuity both during and after the test. Loss of continuity for a period of longer than 10 microseconds constitutes a failure. All wiring shall be in accordance with 4.6.1.
d. When specified (see 3.1), mating probe to be held in position with a retaining nut.



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| Probe | A dia |
|---|---|
| .040 | .041<br>.039 |
| .080 | .081<br>.079 |




WEIGHT
MATERIAL: 1/2 HARD BRASS
(OPTIONAL)

| Spacer size | C | B Dia | | |
|---|---|---|---|---|
| | | Non | Min | Max |
| 1 | .376 | .040 | .0396 | .0416 |
| | .374 | .080 | .0796 | .0816 |
| 2 | .314 | .040 | .0396 | .0416 |
| | .312 | .080 | .0796 | .0816 |
| 3 | .251 | .040 | .0396 | .0416 |
| | .241 | .080 | .0796 | .0816 |

| Inches | mm | Inches | mm |
|---|---|---|---|
| .005 | 0.13 | .252 | 6.40 |
| .016 | 0.41 | .254 | 6.45 |
| .039 | 0.99 | .312 | 7.92 |
| .0396 | 1.01 | .314 | 7.98 |
| .040 | 1.02 | .374 | 9.50 |
| .041 | 1.04 | .376 | 9.55 |
| .0416 | 1.06 | .499 | 12.67 |
| .079 | 2.01 | .501 | 12.73 |
| .0796 | 2.02 | .588 | 14.94 |
| .080 | 2.03 | .874 | 22.20 |
| .081 | 2.06 | .875 | 22.23 |
| .0816 | 2.07 | .876 | 22.25 |
| .241 | 6.12 | 3.98 | 101.09 |
| .249 | 6.32 | 4.02 | 102.11 |
| .251 | 6.38 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents to the nearest .01 mm are given for general information only, and are based on 1 inch =25.4 mm.
3. Unless otherwise specified, tolerance is .XXX ± .005.
4. Break all sharp edges.

FIGURE 3. Test fixture for resistance to test probe damage.



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4.6.11 Vibration (see 3.5.9). Connectors shall be tested in accordance with EIA/ECA-364-38. The following details shall apply:

a. Test condition number - III.
b. Measurement during test - Connectors shall be monitored for electrical continuity during the test. Equipment shall be sufficiently sensitive to detect any interruption with a duration of 10 microseconds.
c. Tests after vibration - Insertion and withdrawal and contact resistance shall be measured as specified in 4.6.7 and 4.6.4 respectively.

4.6.12 Humidity (steady state) (see 3.5.10). Connectors shall be tested in accordance with EIA/ECA-364-31. The following details shall apply:

a. Test-condition letter - B.
b. Final measurements - Insulation resistance shall be measured as specified in 4.6.5.

4.6.13 Salt spray (corrosion) (see 3.5.11). Connectors shall be tested in accordance with EIA/ECA-364-26. The following details shall apply:

a. Test condition letter - B.
b. Measurements after exposure - Immediately after expose, connectors shall be washed with tap water and allowed to dry for 6 hours maximum in a circulating air oven at a temperature of 38°C ± 3°C. After drying, contact resistance shall be measured as specified in 4.6.4.

4.6.14 Durability (see 3.5.12). A test probe conforming to 4.6.1.1, shall be fully inserted into the connector socket contacts and withdrawn for a total of 500 insertion-withdrawal cycles, except as specified (see 3.1), at a rate not to exceed 100 cycles per hour. After the test, insertion and withdrawal and contact resistance shall be measured as specified in 4.6.7 and 4.6.4 respectively.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The test point connectors covered by this specification are primarily for use in airborne, ground support, and shipboard electrical and electronic equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Packaging requirements (see 5.1).
c. Title, number and date of the applicable specification sheet.
d. The complete PIN of the part ordered.
e. Refer to the contract or purchase order for guidance on the disposition of rejected components.
f. When first article inspection is required (see 3.2).

6.3 Subject term (key word) listing.

Closed-entry
Connector
Copper-beryllium
Multiple-contact
Single-Contact

6.4 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy - EC
Air Force – 85
DLA - CC

Review activities:
Army - EA

Preparing activity:
DLA - CC

(Project 5935-2020-008)

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